AI-POWERED HR RESUME SCREENING & RANKING SYSTEM (SMARTFILTER FOR HR)

RECRUIT.AI

Sorted resumes. Smarter hiring.

By -

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PROBLEM STATEMENT

* Recruitment involves handling large volumes of resumes manually
* Traditional systems rely on keyword matching, not understanding context
* High chances of overlooking qualified candidates
* Time-consuming and inefficient process for HR teams
* Lack of automation leads to slower hiring decisions

"Existing systems fail to understand skills semantically."

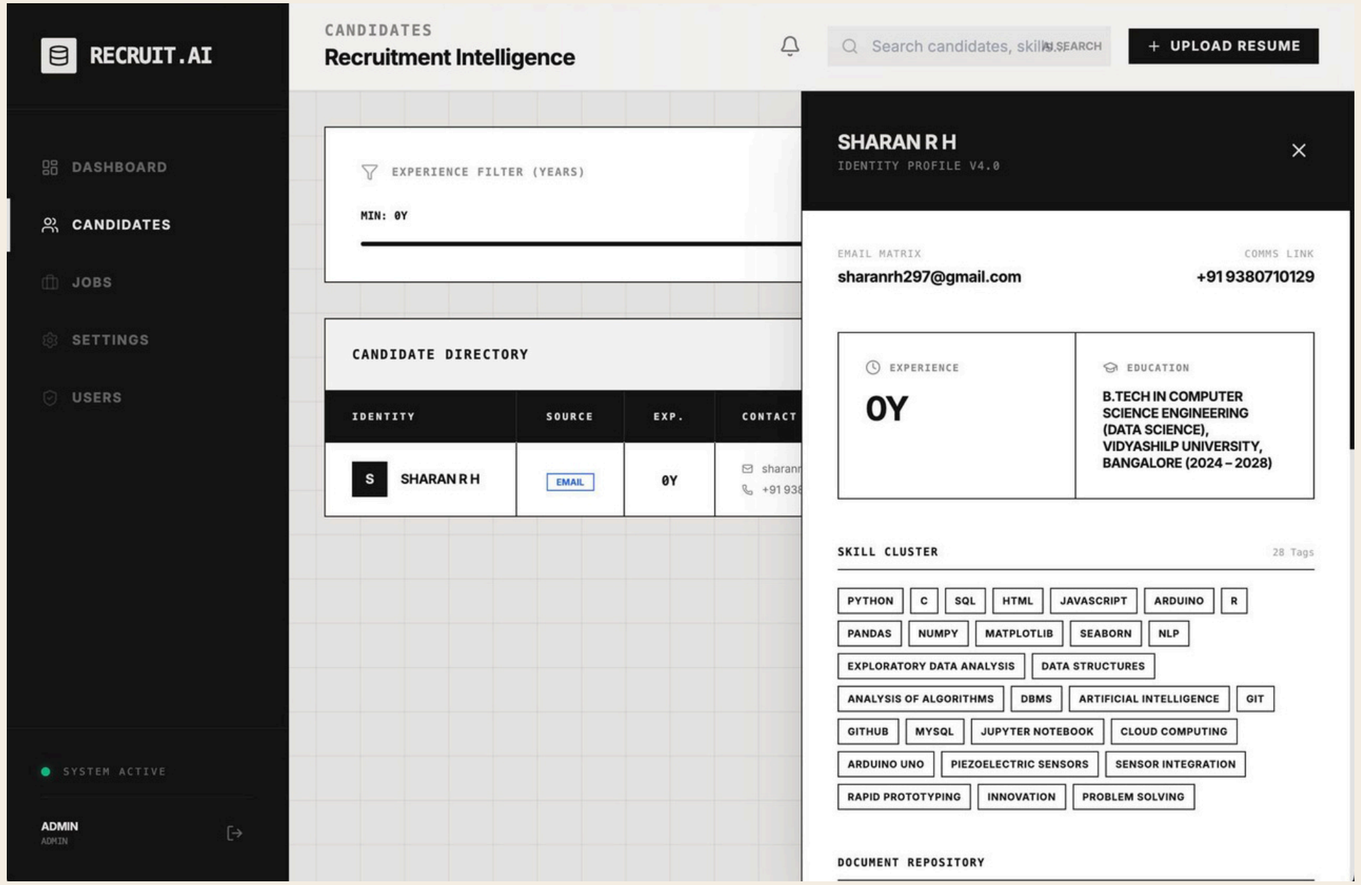
OVERALL SOLUTION DIAGRAM INPUT: RESUME (PDF/DOCX) + JOB DESCRIPTION

Processing:

* AI extracts structured data
* Converts data into embeddings

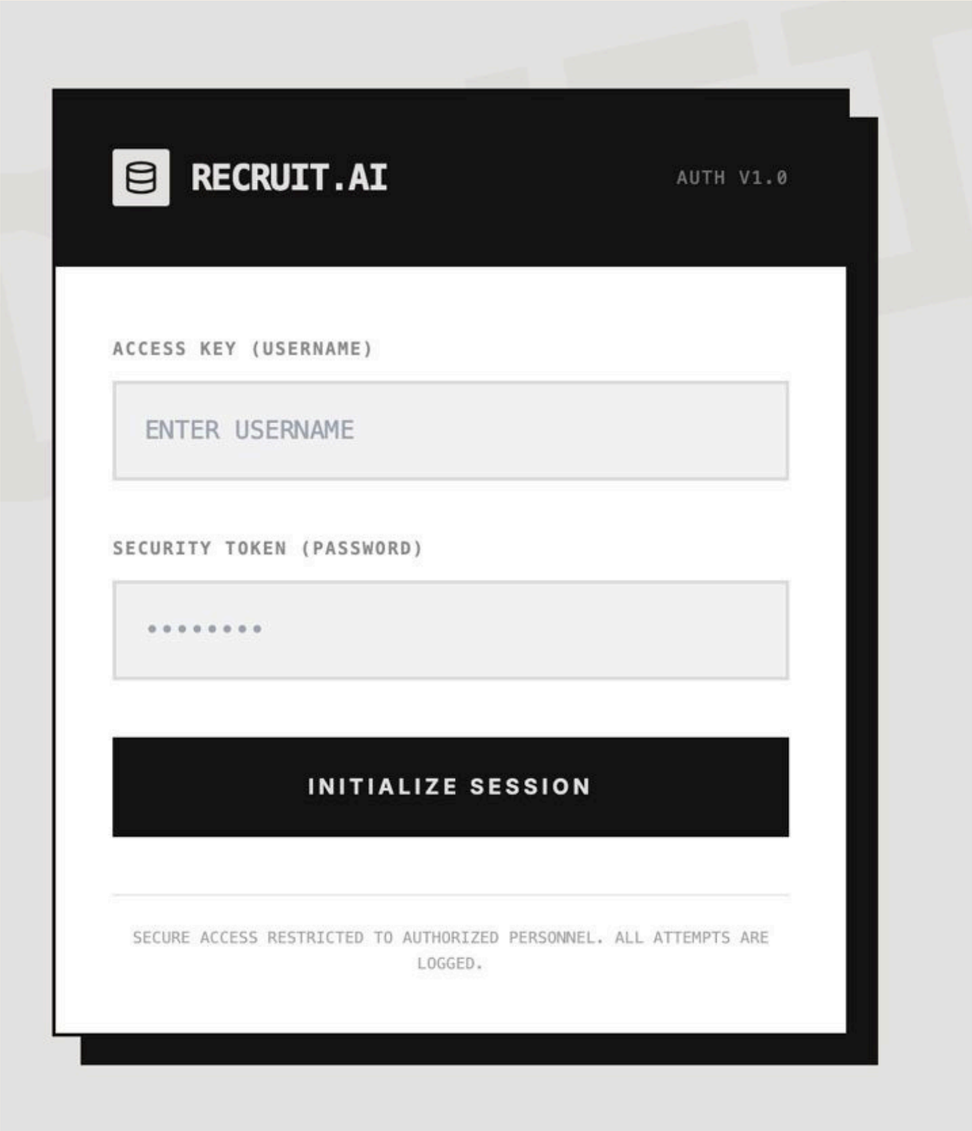
Output:

* Match score
* Ranked candidates
* Dashboard visualization



APPROACH & INNOVATION

- Built a fully automated resume ingestion system
- Used AI to convert unstructured resumes into structured JSON
- Implemented semantic search instead of keyword matching
- Designed modular backend architecture (FastAPI)
- Integrated real-time updates using WebSockets



TECH STACK

Backend:

* Python (FastAPI framework)
* REST API architecture

Frontend:

* HTML, Tailwind CSS, JavaScript

Database:

* SQLite for structured data
* Pinecone for vector storage
* Supabase

AI Tools:

* Google Gemini API for parsing & embeddings

LLM & 5-STEP FRAMEWORK

LLM Used:

- Google Gemini 1.5 Flash

5-Step Framework:

1. Input:
   Resume / Job Description

2. Processing:
   AI extracts key information

3. Structuring:
   Converts into JSON format

4. Embedding:
   Text → Vector representation

5. Matching:
   Calculates similarity score

"This enables understanding beyond keywords."

CHALLENGES FACED

- Handling multiple resume formats (PDF, DOCX)
- Extracting accurate data from unstructured text
- Managing API latency and responses
- Backend-frontend communication issues
- Debugging real-time features (WebSockets)

INDIVIDUAL CONTRIBUTIONS

Backend Development:

- Sharan R H, Mallesha S

Database & Integration:

- Gnanendra Reddy A, Pavan Kalyan M

Frontend & UI Design:

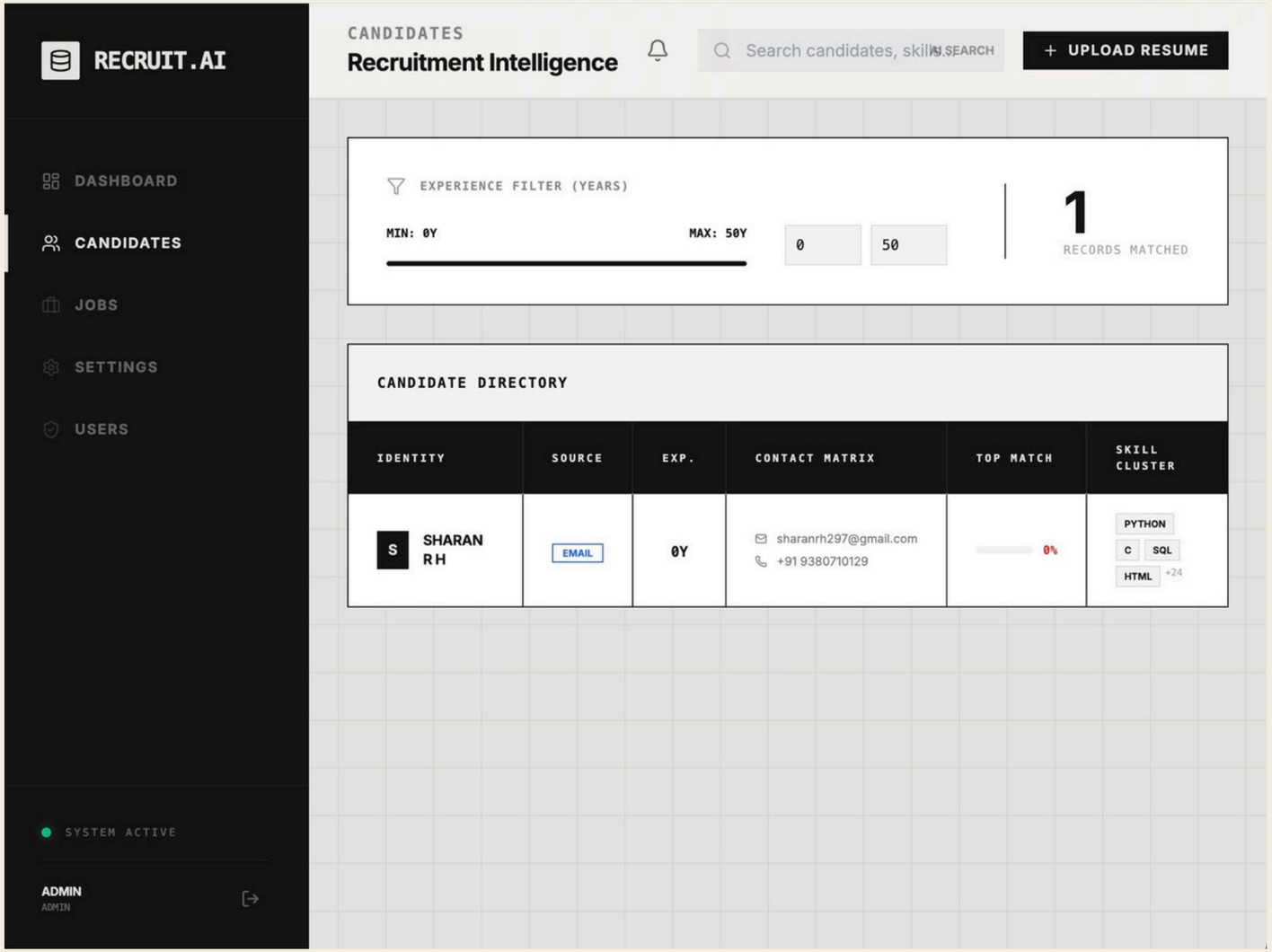
- Venkatesh K, Prajwal M A

Documentation & PPT:

- All team members collaboratively

IMPACT OF SOLUTION

- Reduces manual effort in resume screening
- Speeds up hiring process significantly
- Improves accuracy using semantic matching
- Provides data-driven decision making
- Scalable for real-world recruitment systems



FUTURE OUTLOOK

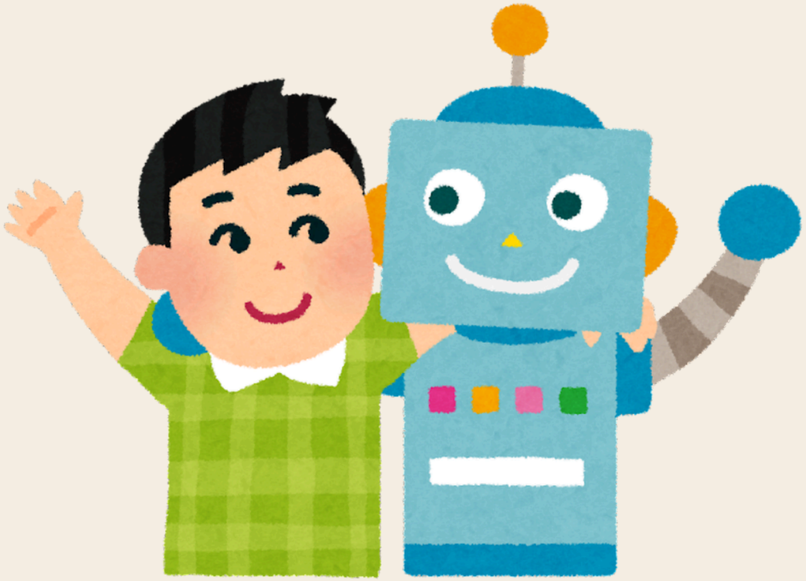
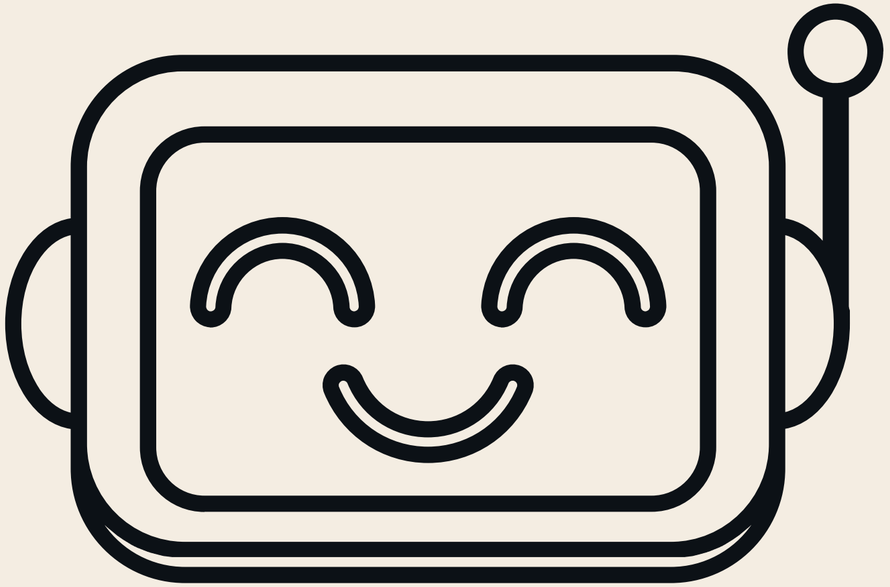
* Automated interview scheduling system
* Integration with LinkedIn and job portals
* Bias detection for fair hiring
* Advanced ML-based recommendation engine
* Candidate feedback and tracking system

CONCLUSION

* AI transforms traditional recruitment into intelligent automation

* System improves efficiency, accuracy, and scalability

* Demonstrates real-world application of AI in HR tech



THANK YOU!